StockEvent Lens — Batch 016 Intelligence Dashboard

Articles 076–080 Analysis, Vector Embedding Sensitivity & Conflict Thresholds

Batch ID
`batch_16 (Articles 076-080)`

TEST 076
Original — 100% Base
Super Micro starts shipping NVIDIA Vera Rubin NVL72 racks (72 GPUs / 36 CPUs).

TEST 077
Repeat — 95% Overlap
Synthetic report restating Vera Rubin NVL72 liquid-cooled rack shipments with zero new facts.

TEST 078
Update — +28% New Info
Commercial customer completes initial acceptance testing on installed Vera Rubin rack.

TEST 079
Conflict — Dispute Specs
Contradicts hardware specs: claims racks contain 128 Rubin GPUs and 64 Vera CPUs.

TEST 080
Different — 0% Overlap
Unrelated weekly Wall Street review (AMD $1T, Meta Muse, Treasury 5.2%) starting Group 032.

Technical Analysis: Dense Vector Blindness vs. Hardware Metric Contradiction

In hardware infrastructure and server aggregation, dense embedding models (e.g., BAAI/bge-small-en-v1.5, OpenAI Ada-002) frequently display critical blindness toward precise numerical configurations. Changing component ratios (e.g., 72 GPUs / 36 CPUs in test_076 vs 128 GPUs / 64 CPUs in test_079) yields high vector similarity (~0.91) because both texts share identical surrounding context tokens ("Super Micro", "NVIDIA Vera Rubin NVL72", "liquid cooling", "rack").

```
// Formula: Formal Hardware Conflict Detection Condition
Conflict_State = ( Cosine_Similarity(E_orig, E_cand) > τ_sim ) && ( ΔF_contradiction == 1.0 )
where τ_sim = 0.60, and ΔF_contradiction = 1.0 indicates a hard hardware parameter or GPU/CPU count dispute.
```

1. Dense Embedding Blindness
Dense embeddings map "72 GPUs" and "128 GPUs" into nearly indistinguishable vector spaces when context terms match. Thus, pure cosine distance completely misses critical hardware specification conflicts.

2. LLM Evaluation Override
To prevent corrupting system architecture specs, the LLM evaluation pipeline performs explicit numerical attribute-value verification, triggering a conflict override when $\Delta F_{contradiction} = 1.0$ is flagged.

Article Relationship & Percentage Breakdown (Batch 016)

FILE IDENTIFIER	PATTERN LABEL	KEYWORDS IDENTIFIED	SIMILARITY %	DELTA %	CLASSIFICATION RATIONALE
`test_076_original`	Original (%5=1)	Super Micro, NVIDIA Vera Rubin NVL72, liquid cooling, 72 GPUs / 36 CPUs, $60B order book	100%	Baseline	Establishes breaking news frame on Super Micro shipping Vera Rubin liquid-cooled rack systems.
`test_077_repeat`	Repeat (%5=2)	Vera Rubin NVL72, direct liquid cooling, 72 Rubin GPUs, 36 Vera CPUs, Building Block Solutions	95%	0% New Facts	Synthetic restatement of Vera Rubin NVL72 rack shipments and identical hardware architecture.
`test_078_update`	Update (%5=3)	**Initial acceptance testing completed**, customer sign-off, installed rack validation	82%	+28% Operational Facts	Advances event state as commercial customer completes acceptance testing on installed rack.
`test_079_conflict`	Conflict (%5=4)	**128 Rubin GPUs / 64 Vera CPUs dispute**, conflicting specs, competing report	91% Match	$\Delta F_{contradiction}$ = 1.0	Directly disputes original hardware specs by asserting 128 GPUs and 64 CPUs per rack.
`test_080_different`	Different (%5=0)	AMD $1T market cap, Meta Muse AI agent, Moderna vaccine rally, 10-Yr Treasury 5.2%	0%	Group 032 Trigger	Zero domain overlap; triggers standalone Group 032 market roundup cluster.

Group 031 — Consolidated Event Summary

4 Articles (076–079)

Super Micro Computer (SMCI) initiated shipments of direct liquid-cooled NVIDIA Vera Rubin NVL72 racks featuring Data Center Building Block Solutions, gigawatt-scale architecture, and a $60B order book (Articles 076–077). Operational progress reached a new milestone as a commercial **customer completed initial acceptance testing on an installed Vera Rubin rack** , validating cooling and networking integration (Article 078). However, hardware specifications are in direct conflict: original reports specified 72 Rubin GPUs and 36 Vera CPUs per rack, whereas a contradictory report claims each **rack contains 128 Rubin GPUs and 64 Vera CPUs** (Article 079 ; $\Delta F_{contradiction}$ = 1.0).

✓ Update Preserved: Customer Acceptance Testing Signed Off
✓ Conflict Preserved: 72 GPU/36 CPU vs 128 GPU/64 CPU Discrepancy
✓ Duplicate Removed: Article 077 Consolidated

Group 032 — Standalone Summary

1 Article (080)

A comprehensive weekly Wall Street review highlighted AMD crossing a $1 trillion market cap , Meta's Muse AI agent triggering an "agentic cancellation wave" in subscription stocks, Moderna's 30% vaccine rally, and 10-year Treasury yields spiking to 5.2% following strong PMI data (Article 080).

✓ New Event Cluster Established ✓ Zero Entity Contamination from Group 031

Batch 016 Raw Data Source Inspector

JSON Format Verification & Evidence Code Snippets

`test_076_original.json` — Group 031

```
{
  "id": 61992909,
  "headline": "What's Going On With Super Micro Comput
  "symbols": ["ARTY", "DELL", "ENTR", "HPE", "IDGT", "I
  "summary": "SMCI stock rises premarket as Super Micr
}
```

Establishes 72 GPU / 36 CPU Vera Rubin baseline.

`test_078_update.json` — Group 031

```
{
  "id": 990000078,
  "headline": "Super Micro completes first gigawatt-sc
  "symbols": ["NVDA", "SMCI"],
  "summary": "Following initial shipments, Super Micro
}
```

Adds customer acceptance testing milestone.

`test_079_conflict.json` — Group 031

```
{
  "id": 990000079,
  "headline": "Super Micro starts shipping NVIDIA Vera
  "symbols": ["NVDA", "SMCI"],
  "summary": "Super Micro began shipping NVIDIA Vera R
}
```

Disputes 72/36 vs 128/64 hardware spec layout.